MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

B.Tech V – Regular End Semester Examination December 2023

SUBJECT: NATURAL LANGUAGE PROCESSING [DSE3155]

Date: 08-12-2023

MARKS: 50 **Duration: 3 Hrs**

Q.NO	Questions	Marks	Course Outcome	Program Outcome	Bloom's Taxonomy
Q1.	a) Construct a NFA (non-deterministic finite automaton) and provide its transition table to recognize strings consisting of 0s and 1s. These strings should either conclude with '010' and have '011' somewhere before it, or conclude with '101' and have '100' somewhere before it in the string. (b) Construct a deterministic finite automaton (DFA) and provide its transition table for a language that recognizes strings in which the occurrence of '0' is directly preceded and followed by '1', over the input alphabets ∑ = {0, 1}.	(5)	CO1	PO1,PO2	L3
Q2.	Construct the regular expressions for the following criteria: a) Strings that either start with 0 and have an odd length or start with 1 and have an even length. b) Strings that begin with a vowel (a, e, i, o, u), followed by any combination of characters from a to z (including zero occurrences), and conclude with a consonant (any letter that is not a vowel). c) Strings that represent prices in the format of $3.45, $23.32, or $400."	(3)	CO1	PO2,PO1	L3
Q3.	Outline the inherent differences between Non-Deterministic Finite Automata and Deterministic Automata with suitable justifications	(2)	CO1	PO1,PO2	L2

Q4.	Apply Byte-Pair encoding for the given text corpus with the following four words: “ab”, “bc”, “bcd”, and “cde”. Provide insights into how the byte pair encoding contributes to compressing the data.	(5)	CO2	PO1,PO2	L3
Q5.	Explain the fundamental elements crucial for building a morphological parser. Substantiate your discussion with specific examples in the development and functionality of morphological parsers	(3)	CO2	PO1	L2
Q6.	Identify the inflectional and derivational morphemes present. In the following sentence. ***“The cat's playful antics often amuse the family. Despite its small size, the kitten displays a remarkable level of intelligence”.***	(2)	CO2	PO1	L3
Q7.	Compute the minimum edit distance between the words "sunshine" and "moonlight" using the bottom-up dynamic programming approach. Trace the algorithm and list the type of edit operations that need to be performed	(5)	CO3	PO1,PO2	L3
Q8.	Compute the Maximum likelihood estimation of the test sequence **P(<S>They play in a big garden<\S>)** using bi-gram model in the given training corpus. <S> There is a big garden</S> <S> Children play in the garden</S> <S> They play inside beautiful garden </S> <S> Trees in a garden </S>. Construct the uni-gram count, bi-gram count, normalized bi-gram count tables for maximum likelihood estimation.	(3)	CO3	PO2	L3
Q9.	Evaluate the effectiveness of the Noisy Channel Model in correcting errors and preserving the integrity of information.	(2)	CO3	PO1,PO2	L4
Q10.	Given a corpus and three parts-of-speech (POS) tags - noun (N), adjective (Adj), verb (V) and Determiner (Det), Apply the Stochastic tagging algorithm to compute the probabilities for each	(5)	CO3	PO1	L3

	sentence and tag each word accordingly. Corpus: “The cat is sleeping. Sleeping cats are cute. Cats can sleep all day. Sleepy cats often yawn”.				
Q11.	Summarize the situations one needs to address to convert the grammar to Chomsky-Normal form	(3)	CO4	PO1,PO2	L2
Q12.	Elaborate on the challenges associated with parts-of-speech tagging (POS tagging)	(2)	CO4	PO1	L2
Q13.	Apply probabilistic CYK parser using the given PCFG rules for the sentence “The cat chased the mouse”. [The probability of each production is given in []). S -> NP VP [0.4] NP -> Det N [0.3] NP -> NP PP [0.1] VP -> V NP [0.2] PP -> P NP [0.5] Det -> "the" [0.4] | "a" [0.3] | "an" [0.3] N -> "cat" [0.3] | "dog" [0.4] | "mouse" [0.3] V -> "chased" [0.2] | "caught" [0.4] | "saw" [0.4] P -> "with" [0.6] | "on" [0.2] | "under" [0.2] b.) Given the sentence "**kids saw birds with fish**", build all possible parse trees for the sentence and compute the probability of each parse tree according to the provided probabilities for each production rule. Finally, determine the most probable parse tree for the given sentence. [The probability of each production is given in []). S → NP VP [1.0] NP → NP PP [0.4] PP → P NP [1.0] NP → kids [0.1] VP → V NP [0.7] NP → birds [0.18] VP → VP PP [0.3] NP → saw [0.04] P → with [1.0] NP → fish [0.18] V → saw [1.0] NP → binoculars. [0.1].	(5)	CO4	PO5	L3
Q14.	Given the below production rules:	(3)	CO4	PO1,PO5	L4

	S → NP VP S → S CC S NP → Pronoun | DET Nominal Nominal → Noun | ADJ Nominal | DET ADJ Nominal VP → V NP VP → V PP PP → P NP DET → the | a | an Noun → car | house | cat | tree ADJ → red | big | fluffy V → runs | eats | sleeps P → on | under | beside CC → and | or i. Build the parse tree using top-down parsing for the sentence: ***"The big red car runs under the tree and the fluffy cat sleeps on the house"*** ii. Identify the distinct characteristics of the top-down and bottom-up parsing strategies in the context of generating the parse trees				
Q15.	Explain the various sentence-level constructions for English sentences with suitable examples.	(2)	CO4	PO1,PO5	L2